Solar ESS Co., Ltd.
53 Moo 2, Nang Lae, Mueang, Chiang Rai 57100, Thailand
Tel. 063 151 6694, 089 632 5829 · LINE: solaress1
www.solaress.org · www.โซลาร์เอสส์.com

DC combiner set compared: fuse, DC breaker and DC SPD

Solar ESS product comparison · October 2026 · Suntree, FEEO, CHINT, Schneider, ABB, TOMZN

> **Our pick: Suntree.** The only brand with all three parts — fuse and holder, DC breaker and DC SPD — readily available in Thailand, with TUV/CE/CB marks stated. ABB has the best engineering margin but costs about ten times more and is hard to obtain in Thailand.

Brand	Fuse + holder (10×38, 1000 V)	DC breaker	DC SPD Type 2, 1000 V	Relative set cost	Notes
Suntree	SRD-30 holder + fuse	SL7N-63 2P 800 V (4P 1000 V available)	SUP2H1-PV 3P	Low	IEC/EN 60947-2, TUV/CE/CB stated; good Thai stock
FEEO (budget)	FDS-32 + 25 A fuse	FPV-63 4P 1000 V	FSP-D40 3P 1000 V	Lowest	No certificates cited in listings
CHINT	No genuine PV fuse found in Thailand	NB1-63DC 4P 1000 V	NU6-II PV: no Thai listing	Medium	Grey-market in Thailand
Schneider	No Thai listing	C60PV-DC 2P 800 V	iPRD-DC: no Thai listing	High	Expensive
ABB	E90 PV holder	S802PV-S 2P 800 V	OVR PV	Very high	Best margin, poor Thai stock
TOMZN	Generic	TOB7Z-63 2P '1000 V'	TZG40-PV	Lowest	2P 1000 V claim doubtful; avoid

800 V or 1000 V breaker?

Most 2-pole DC breakers from major brands are rated 800 V DC. That is enough when the string's open-circuit voltage on the coldest morning stays below 800 V, which is typical for 10 kW hybrids with 12–14 panels per string. Above 800 V, use a 4-pole breaker rated 1000 V, with the poles wired in series. Fuse rating must be at least 1.56 × the panel's Isc and no more than the panel's maximum series fuse.

Specifications are taken from the manufacturers' published datasheets; check the current datasheet before ordering. For prices and a design for your site, book a consultation at www.solaress.org or LINE: solaress1.

Sources: Suntree SL7N · Schneider C60PV-DC · ABB S800PV